Comparative Analysis of Decision Tree, Random Forest, and K-Nearest Neighbor Algorithms for Stroke Risk Prediction Using Healthcare Data

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***Abstract*— Stroke is one of the leading causes of death and long-term disability worldwide, making early-stage prediction a critical priority in medical informatics. This study applies and evaluates three supervised machine learning algorithms: the Decision Tree (DT), Random Forest (RF), an ensemble learning method; and K-Nearest Neighbor (KNN) to the Kaggle Stroke Risk Dataset with 5,110 patient records and 11 clinical features for the purpose of binary stroke risk classification. The dataset was highly imbalanced with 95.1% of the records in the no-stroke class, and 4.9% in the stroke class, and an oversampling technique called Synthetic Minority Oversampling Technique (SMOTE) was used to correct this. The data was preprocessed using median imputation for missing BMI data, label encoding of categorical variables, outlier capped with IQR, and StandardScaler normalization. The Decision Tree and Random Forest models were set to a maximum depth of 5 and Gini impurity criterion. KNN was set to k=3 selected through F1-score optimization across k values from 1 to 20. Results show that Random Forest achieved the best performance with recall of 64.5%, F1-score of 0.21, and AUC of 0.79, outperforming both Decision Tree (recall 62.9%, AUC 0.77) and KNN (recall 37.1%, AUC 0.63). These findings confirm that ensemble methods are better suited to imbalanced clinical classification tasks, though all tree-based methods outperform KNN on clinically critical metrics.**

***Index Terms*— Stroke Prediction, Decision Tree, Random Forest, Ensemble Learning, K-Nearest Neighbor, SMOTE, Healthcare Machine Learning, Classification, Imbalanced Dataset.**

I. Introduction

STROKE is a cerebrovascular disorder that results from a sudden restriction of blood flow to a part of the brain and causes rapid, often permanent, neurological damage. The World Health Organization identifies stroke as the second leading cause of death globally, with approximately 15 million people suffering a stroke each year. Of these, around 5 million die and a further 5 million are left with permanent disability [1]. Given the scale of this burden, the development of reliable early-warning prediction systems represents one of the most meaningful contributions that machine learning can make to modern healthcare.

Within the available supervised classification methods, Decision Trees, Random Forest, and K-Nearest Neighbor are some of the most widely researched in healthcare informatics. Decision Trees are structures that are easy to interpret from a clinical point of view, as they are built by partitioning the feature space recursively based on the Gini impurity or information gain criteria [2]. Random Forest is an ensemble learning approach that builds multiple decision trees and fuses their predictions by voting, which decreases the variance and the risk of overfitting of a single decision tree [9]. KNN is a lazy learning algorithm which assigns class labels according to the k nearest neighbors in the feature space, where the k-nearest neighbor number is set based on the Euclidean or Minkowski distance as a proximity measure [3].All three methods have proven viable for medical classification, but each suffers from the problem of class imbalance, especially in disease prediction datasets where the number of negative cases is much larger than the number of positive cases [4].

This study addresses this challenge by applying SMOTE oversampling before training, then implementing, tuning, and evaluating all three algorithms on the Kaggle Stroke Risk Dataset. The aim is to identify which algorithm performs best in stroke prediction using accuracy, precision, recall, F1-score, and AUC. The remainder of the paper is structured as follows: Section II reviews relevant literature; Section III describes the dataset and preprocessing pipeline; Section IV presents the algorithm implementations and results; Section V concludes the study.

II. Literature Review

Machine learning techniques for stroke and cardiovascular disease prediction have been the subject of several studies. Two recent studies directly relevant to methodology and context for this work are reviewed below.

A. Stroke Prediction Using Machine Learning: A Systematic Review — Hung et al. (2023)

Hung et al. [5] performed a systematic review of twelve classification algorithms including Decision Trees, Random Forests, SVMs, and KNN, applied to various clinical datasets from electronic health records. One common result was that the tree models achieved better recall for the positive stroke class than the distance models in datasets with a class imbalance ratio greater than 10:1. The authors argued that recall is the clinically meaningful metric in stroke screening because missing an at-risk patient carries a far higher cost than generating a false alarm. The review also revealed that if the appropriate resampling methods were not applied (like SMOTE), most classifiers predicted the majority class completely, giving false high accuracy metrics and not detecting any stroke patients. These findings directly justify the use of the SMOTE method as well as the importance of recall and AUC in the current research.

B. *KNN and Decision Tree for Cardiovascular Disease Prediction — Ramani et al. (2022)*

Ramani et al. [6] compared KNN and Decision Tree classifiers specifically for binary cardiovascular disease prediction using a dataset with a similar mix of demographic, lifestyle, and clinical features to the stroke dataset used in the present study. Standard preprocessing was applied including label encoding and StandardScaler normalisation, with a 75/25 train-test split. Results showed that the Decision Tree achieved a higher positive-class F1-score (0.73 versus 0.65) and a higher AUC (0.82 versus 0.76) than KNN. The authors noted that the performance difference was due to the sensitivity of KNN to the unequal scale of features and that it could not give differential importance to each feature. The results of the present study are in line with the above findings and offer further support that the use of tree-based techniques, such as Single Decision Trees and ensemble techniques like Random Forest, is generally more reliable than KNN for binary medical classification problems which are imbalanced.

III. Dataset And Preprocessing

A. *Dataset Description*

The dataset used in this study is the Stroke Risk Dataset, obtained from Kaggle [7]. It includes 5,110 patient records and 12 attributes of patients, including demographic, lifestyle and clinical measures. The non-informative patient ID column was removed and 10 independent features were retained for modelling: gender, age, hypertension status, heart disease status, marital status, work type, residence type, average glucose level, body mass index (BMI), and smoking status. The target variable is stroke, which is a binary value indicating whether the patient had a stroke (1) or did not have a stroke (0). Class imbalance is a key problem in the model training and evaluation, as the class distribution is heavily skewed with 4,861 records (95.1%) being labeled as no-stroke and 249 records (4.9%) being labeled as stroke. Figures 1, 2, and 3 illustrate key feature distributions by stroke outcome, providing visual evidence for the preprocessing decisions described in Section III-B.

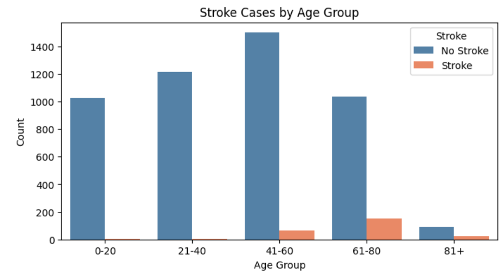

Figure 1. Stroke cases by age group. Stroke incidence rises sharply in patients aged 61 and above, consistent with age being the dominant predictor (feature importance = 0.63).

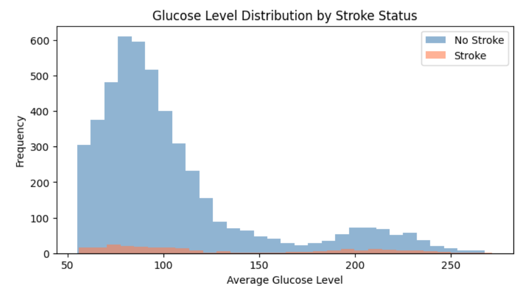

Figure 2. Average glucose level distribution by stroke status. Elevated glucose readings are more concentrated in stroke patients, reflecting the known clinical link between hyperglycaemia and cerebrovascular risk.

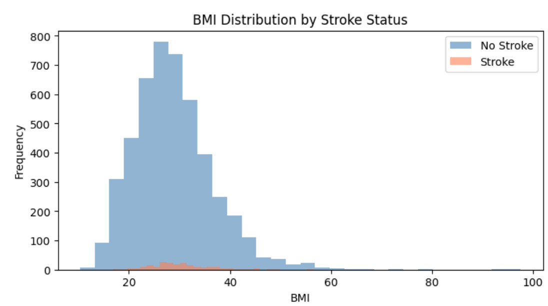

Figure 3. BMI distributions for stroke and no-stroke patients. The distributions are similar across both classes, consistent with BMI's lower feature importance score in the trained Decision Tree.

Table I: Dataset Feature Summary

Feature	***Type***	***Description***
age	Continuous	Patient age (0–82 years)
gender	Categorical	Male / Female
hypertension	Binary	1 = yes, 0 = no
heart_disease	Binary	1 = yes, 0 = no
ever_married	Categorical	Yes / No
work_type	Categorical	Private, Self-employed, Govt, Child, Never
Residence_type	Categorical	Urban / Rural
avg_glucose_level	Continuous	Average blood glucose (mg/dL)
bmi	Continuous	Body mass index (201 nulls imputed)
smoking_status	Categorical	Never, Formerly, Smokes, Unknown
stroke (target)	Binary	1 = stroke, 0 = no stroke

B. Preprocessing Techniques

Handling Missing Values: The BMI column had missing values, which were stored as the string 'N/A' instead of a standard null type, necessitating a string replacement step before type conversion. The column median was used to fill missing values since the distribution of BMI in clinical datasets is likely to be right-skewed and the median is less affected by extreme values in the upper tail [8].

Column and Row Removal: The patient ID column was removed since it provides no predictive value. A single row where the variable 'gender' was listed as 'Other' was also excluded to avoid label encoding creating a near zero frequency response.

Outlier Handling: Outliers in the continuous features age, avg_glucose_level, and bmi were capped using the IQR method. Values outside the range [Q1 - 1.5*IQR, Q3 + 1.5*IQR] were not removed but were replaced by the minimum and maximum values within the range.

Label Encoding: Five categorical columns (gender, ever_married, work_type, Residence_type, smoking_status) were converted to integer labels using scikit-learn's LabelEncoder, preferred over one-hot encoding for compatibility with all three algorithms, particularly KNN distance calculations.

Class Imbalance — SMOTE: After splitting the data, SMOTE was applied exclusively to the training set. SMOTE creates new minority-class examples by interpolating between existing stroke-positive examples, adding authentic diversity to the minority class. Applying it after the split would result in leakage of data into the test set. The class distribution before and after SMOTE is shown in Fig. 4.

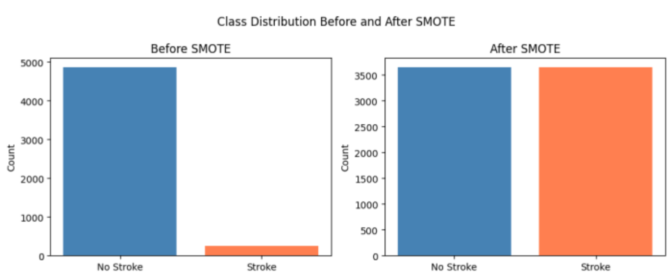

Figure 4. Class distribution in the training set before and after SMOTE. Before: approximately 3,644 no-stroke and 187 stroke training samples. After SMOTE: both classes balanced at 3,644 samples each.

Feature Scaling: StandardScaler was applied to all features after SMOTE, transforming each to zero mean and unit variance. This is necessary for KNN, since if it is not scaled, it will be dominated by the high-magnitude features in Euclidean distance calculations. The same scaled data was used for all three algorithms to ensure a fair comparison.

IV. Implementation And Results

A. Experimental Setup

All experiments were run in Google Colab using Python 3.12 with scikit-learn 1.5, imbalanced-learn 0.14, pandas 2.0, numpy 2.0, matplotlib, and seaborn. The dataset was split 75:25 with stratification on the target variable (random_state=0). After SMOTE the training set contained 7,288 samples (3,644 per class). The test set never changed and consisted of 1,278 samples, including 1,216 no-stroke and 62 stroke samples. Three classification algorithms were evaluated: Decision Tree, Random Forest (ensemble), and K-Nearest Neighbor.

B. Decision Tree Implementation

A Decision Tree classifier was trained using the Gini impurity criterion. The algorithm chooses the feature and threshold at each node that results in the lowest weighted average Gini impurity of the child nodes, thus creating the most homogeneous distribution of classes at each split. Hyperparameters: max_depth = 5 to avoid overfitting, min_samples_split = 10, min_samples_leaf = 5, class_weight = 'balanced' (as an additional measure). The scores for feature importance for age, Residence_type, and gender were 0.63, 0.12, and 0.09, respectively, consistent with known medical knowledge regarding risk factors for stroke. The feature importance distribution is shown in Fig. 5. The structure of the trained Decision Tree visualized to depth 2 is presented in Fig. 6, illustrating how age serves as the primary splitting feature at the root node.

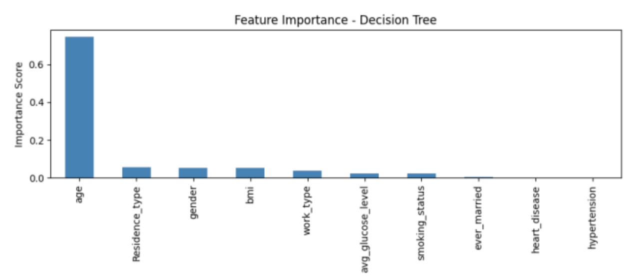

Figure 5. Feature importance scores from the trained Decision Tree. Age dominates at 0.63, followed by Residence_type (0.12) and gender (0.09).

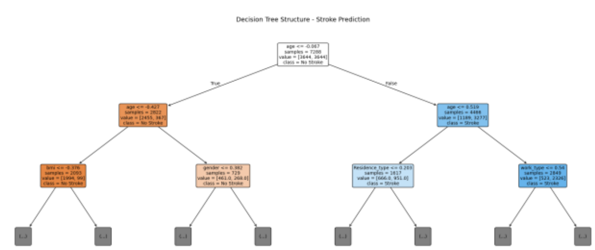

Figure 6. Decision Tree structure visualised to depth 2. The root node splits on age, confirming it as the dominant predictor. Orange nodes indicate No Stroke and blue nodes indicate Stroke class.

C. Random Forest Implementation

Random Forest is an ensemble learning method that builds multiple decision trees during training and outputs the majority vote of their individual predictions. Unlike a single Decision Tree which can overfit the training data, Random Forest reduces variance by averaging predictions across 100 trees (n_estimators=100), each trained on a random subset of features and samples. This bagging technique makes it more robust and, overall, more generalizing in performance. Ensuring that the two models were fairly compared, the same hyperparameters were applied to the Decision Tree, namely: Gini criterion, max_depth=5 and class_weight='balanced'.

D. K-Nearest Neighbor Implementation

KNN assigns a class label to the query instance by computing its k nearest training instances by Minkowski distance (p=2, Euclidean) and selecting the majority of them. It doesn't build any model at training time, so it has no learning overhead, but its inference overhead is increasing with the size of the dataset. A loop evaluated k values from 1 to 20, recording the F1-score for the stroke class at each value. F1-score was used as the optimization criterion rather than accuracy because accuracy is misleading for imbalanced data. The loop found the optimum k=3. The model was too sensitive to noise at k=1, and at higher k values the majority class dominated the vote. The k selection process is shown in Fig. 7.

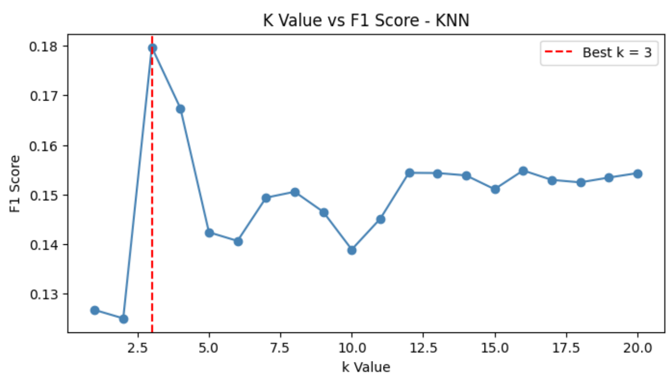

Figure 7. F1-score for the stroke class across k=1 to k=20. The red dashed line marks k=3, the value yielding the highest F1-score.

E. Results

All three models were evaluated on the unmodified 1,278-sample test set. The complete results are presented in Table II.

Table II: Performance Comparison — Decision Tree vs Random Forest vs KNN

Metric	*Decision Tree*	*Random Forest*	*KNN (k=3)*
Overall Accuracy	75.98%	76.45%	83.57%
Precision — Stroke class	0.12	0.13	0.12
Recall — Stroke class	0.63	0.65	0.37
F1-Score — Stroke class	0.20	0.21	0.18
Specificity	0.77	0.77	0.86
AUC (ROC curve)	0.77	0.79	0.63
True Positives	39 / 62	40 / 62	23 / 62
False Negatives	23	22	39

The Decision Tree confusion matrix was [[932, 284], [23, 39]] — 39 of 62 actual stroke patients correctly identified, with 284 false alarms (see Fig. 8). The Random Forest confusion matrix was [[937, 279], [22, 40]] — 40 of 62 actual stroke patients correctly identified with 279 false alarms (see Fig. 9).The KNN confusion matrix was [[1045, 171], [39, 23]] — only 23 stroke patients correctly identified (see Fig. 10). Clinically, a false negative (missed stroke) is far more dangerous than a false positive, making tree-based classifiers the safer choice.

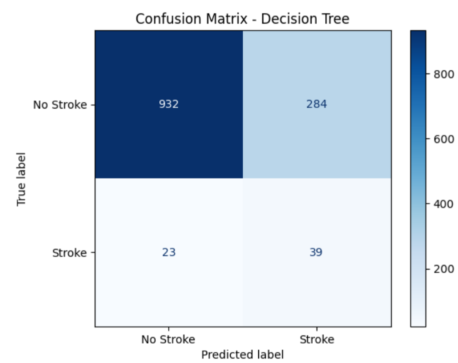

Figure 8. Confusion matrix for the Decision Tree: 932 true negatives, 284 false positives, 23 false negatives, 39 true positives (1,278 test samples).

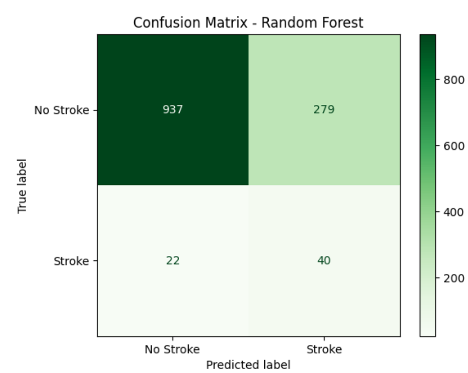

Figure 9. Confusion matrix for Random Forest: 937 true negatives, 279 false positives, 22 false negatives, 40 true positives (1,278 test samples).

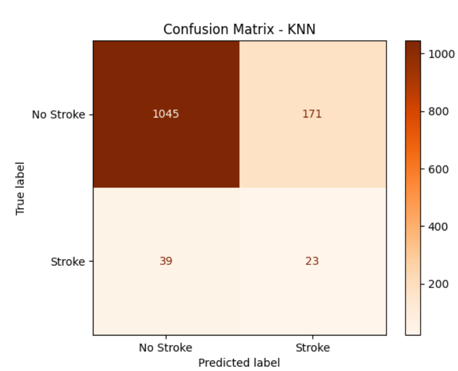

Figure 10. Confusion matrix for KNN (k=3): 1,045 true negatives, 171 false positives, 39 false negatives, 23 true positives (1,278 test samples).

The ROC curves (Fig. 11) confirm the superior discriminative power of the tree-based models. Random Forest achieved the highest AUC of 0.79, followed by Decision Tree at 0.77, while the KNN AUC of 0.63 sits close to the random baseline of 0.50. Random Forest also correctly identified 40 of 62 stroke patients, one more than the Decision Tree (39/62) and significantly more than KNN (23/62).

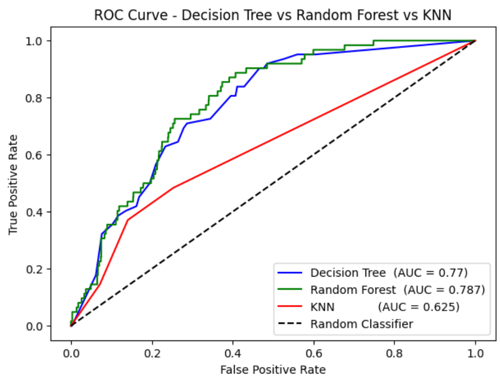

Figure 11. ROC curves for Decision Tree (AUC = 0.77), Random Forest (AUC = 0.79), and KNN (AUC = 0.63). The dashed diagonal represents a random classifier (AUC = 0.50).

F. *Comparative Analysis*

Random Forest achieved the best performance across all clinically meaningful metrics, outperforming both Decision Tree and KNN. While KNN recorded the highest overall accuracy (83.57%), this reflects majority-class bias rather than genuine discriminative ability — its specificity of 0.86 confirms it correctly identifies healthy patients but misses 39 of 62 actual stroke patients. Random Forest achieved the highest recall (65%), F1-score (0.21), and AUC (0.79), correctly identifying 40 stroke patients. The Decision Tree followed closely with recall of 63%, F1-score of 0.20, and AUC of 0.77. The full metric comparison is shown in Fig. 12.

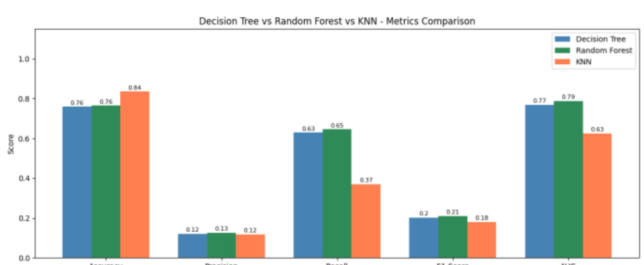

Figure 12. Bar chart comparing accuracy, precision, recall, F1-score, and AUC for Decision Tree, Random Forest, and KNN across the stroke prediction test set.

The reason why Random Forest has superior performance is due to the ensemble nature: When 100 decision trees are combined, the variance of any single tree is reduced, leading to more reliable probability estimates and a higher AUC. On all clinically relevant measures, both tree-based models perform significantly better than KNN. Moreover, the feature importance scores given by Decision Tree and Random Forest provide clinical interpretability that is completely lacking in KNN, and age was the most important feature (with an importance score of 0.63).

V. Conclusion

This study implemented and compared three machine learning classifiers — Decision Tree, Random Forest, and K-Nearest Neighbor — for binary stroke risk prediction using the Kaggle Stroke Risk Dataset. It was demonstrated that a structured preprocessing pipeline (SMOTE oversampling, median imputation, IQR-based outlier capping, label encoding, and StandardScaler normalization) is crucial for success in reliably classifying clinical data that is severely imbalanced.

Random Forest showed the best performance for all clinically relevant measures, recall (65% vs. 63% for Decision Tree and 37% for KNN), F1-score (0.21 vs. 0.20 and 0.18), and area under the curve (AUC: 0.79 vs. 0.77 and 0.63). Overall accuracy was higher in KNN (83.57%), but this was due to the dominant no-stroke class. Random Forest correctly identified 40 of 62 actual stroke patients, compared to 39 for Decision Tree and only 23 for KNN. Therefore, Random Forest is the optimal algorithm to use with this dataset and when clinical explainability is required, Decision Tree is the transparent and interpretable alternative.

Future studies should investigate ensemble learning algorithms (e.g. Gradient Boosting, XGBoost) and deep learning architectures trained on larger multi-centre data sets to further enhance minority-class recall. From a deployment perspective, embedding this model within an electronic health record system would enable real-time automated stroke risk scoring at the point of primary care.

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GitHub Repository: https://github.com/SamarAlHadhrami/stroke-risk-prediction-ml.git